

Important Probability Distributions

1 Discrete Probability Distributions

1.1 Binomial Distribution $\text{Bin}(n, p)$

Models the number of successes in n independent Bernoulli trials.

- **PMF:** $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x \in \{0, 1, \dots, n\}$
- **Mean:** $E(X) = np$
- **Variance:** $\text{Var}(X) = np(1-p)$
- **MGF:** $M_X(t) = (1-p+pe^t)^n$

1.2 Negative Binomial Distribution $\text{NB}(r, p)$

Models the number of failures preceding the r -th success in a sequence of independent Bernoulli trials.

- **PMF:** $f(x) = \binom{x+r-1}{r-1} p^r (1-p)^x$ for $x \in \{0, 1, 2, \dots\}$
- **Mean:** $E(X) = \frac{r(1-p)}{p}$
- **Variance:** $\text{Var}(X) = \frac{r(1-p)}{p^2}$
- **MGF:** $M_X(t) = \left(\frac{p}{1-(1-p)e^t} \right)^r$

1.3 Geometric Distribution $\text{Ge}(p)$

A special case of the Negative Binomial distribution ($r = 1$). It represents the number of failures before the first success.

- **PMF:** $f(x) = p(1-p)^x$ for $x \in \{0, 1, 2, \dots\}$
- **Mean:** $E(X) = \frac{1-p}{p}$
- **Variance:** $\text{Var}(X) = \frac{1-p}{p^2}$
- **MGF:** $M_X(t) = \frac{p}{1-(1-p)e^t}$
- **Key Property:** Possesses the “Lack of Memory” (memoryless) property, meaning $P(X \geq m+n \mid X \geq m) = P(X \geq n)$.

1.4 Poisson Distribution $\text{Po}(\lambda)$

Models the number of times an event occurs in a specific interval of time or space.

- **PMF:** $f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x \in \{0, 1, 2, \dots\}$
- **Mean:** $E(X) = \lambda$
- **Variance:** $\text{Var}(X) = \lambda$
- **MGF:** $M_X(t) = e^{\lambda(e^t-1)}$

1.5 Discrete Uniform Distribution $U(\{1, 2, \dots, N\})$

Models a scenario where all N distinct outcomes are equally likely.

- **PMF:** $f(x) = \frac{1}{N}$ for $x \in \{1, 2, \dots, N\}$
 - **Mean:** $E(X) = \frac{N+1}{2}$
 - **Variance:** $\text{Var}(X) = \frac{N^2-1}{12}$
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2 Continuous Probability Distributions

2.1 Uniform (Rectangular) Distribution $U(\alpha, \beta)$

Models variables where probability is distributed evenly across a continuous interval (α, β) .

- **PDF:** $f(x) = \frac{1}{\beta-\alpha}$ for $\alpha < x < \beta$
- **Mean:** $E(X) = \frac{\alpha+\beta}{2}$
- **Variance:** $\text{Var}(X) = \frac{(\beta-\alpha)^2}{12}$
- **MGF:** $M_X(t) = \frac{e^{\beta t} - e^{\alpha t}}{t(\beta-\alpha)}$

2.2 Gamma Distribution $\text{GAM}(\alpha, \theta)$

Defined by a shape parameter $\alpha > 0$ and a scale parameter $\theta > 0$.

- **PDF:** $f(x) = \frac{1}{\theta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\theta}$ for $x > 0$
- **Mean:** $E(X) = \alpha\theta$
- **Variance:** $\text{Var}(X) = \alpha\theta^2$
- **MGF:** $M_X(t) = (1 - \theta t)^{-\alpha}$

2.3 Exponential Distribution $\text{Exp}(\theta)$

A special case of the Gamma distribution where $\alpha = 1$. Commonly used to model waiting times.

- **PDF:** $f(x) = \frac{1}{\theta} e^{-x/\theta}$ for $x > 0$
- **Mean:** $E(X) = \theta$
- **Variance:** $\text{Var}(X) = \theta^2$
- **MGF:** $M_X(t) = (1 - \theta t)^{-1}$
- **Key Property:** Possesses the memoryless property.

2.4 Normal Distribution $N(\mu, \sigma^2)$

The foundational symmetric, bell-shaped distribution.

- **PDF:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- **Mean:** $E(X) = \mu$
- **Variance:** $\text{Var}(X) = \sigma^2$
- **MGF:** $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$